A Rotational Construction of Genus g Surfaces from $(g-1)$ Octagons
Part 1: Fundamental Construction, Properties, and Further Remarks

Mingli Yuan Gemini ChatGPT DeepSeek

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Abstract

This note details a constructive method for orientable closed surfaces of genus $g \geq 2$. It leverages the concept of a genus g surface S_g as an $N = (g-1)$-sheeted Z_N-symmetric regular covering of a genus 2 surface S_2. We demonstrate that by gluing N octagonal fundamental domains of S_2 according to specific rules derived from this covering structure, a surface of genus g is indeed formed. This approach provides an alternative perspective to the standard construction of S_g from a single $4g$-sided polygon and explicitly highlights the hierarchical relationship between S_g and S_2. We present the general construction, a proof of its resulting genus using Euler characteristics, and illustrate it with a detailed example for $g = 4$.

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1 Introduction: The Special Role of Genus 2 Surfaces and Rotational Constructions

The study of Riemann surfaces and their topological classification is a cornerstone of modern geometry and topology. Closed orientable surfaces are uniquely classified by their genus g, which intuitively counts the number of "handles" on the surface. The standard method for constructing a surface of genus g involves identifying the sides of a single $4g$-sided polygon in a specific sequence (e.g., $a_1 b_1 a_1^{-1} b_1^{-1} \dots a_g b_g a_g^{-1} b_g^{-1}$).

Genus $g = 2$ surfaces hold a special position as they are the simplest (lowest genus) closed hyperbolic surfaces. Their fundamental domain can be represented

by an octagon with sides identified in pairs. Observations of highly symmetric surfaces, such as specific examples of higher-genus surfaces exhibiting rotational symmetry as coverings of lower-genus surfaces (e.g., an 11-hole torus as a 5-fold rotationally symmetric cover of a 3-hole torus), suggest the utility of "rotational construction methods." These methods often involve building a more complex surface from simpler units arranged with specific symmetries, or viewing a surface as a quotient of a more symmetric one by a group action.

This note explores a particular construction method for a genus g surface (S_g) that emerged from considering S_g as an $N = (g-1)$-sheeted regular covering of a fixed genus 2 surface (S_2). The deck transformation group for this covering is the cyclic group Z_N. This perspective leads to a method of building S_g from N octagonal fundamental domains associated with S_2, glued according to rules dictated by the covering structure and the chosen Z_N symmetry. This approach is distinct from the standard single $4g$-gon construction. It offers insights into the hierarchical relationship between surfaces of different genera, with S_2 (represented by its octagon) acting as a foundational building block for these specific symmetric constructions, equivalent to the traditional $4g$-gon description but better highlighting this hierarchical structure. We detail this construction, verify the genus of the resulting surface, and provide a concrete example.

2 The General Construction from $N = (g-1)$ Octagons

Let S_2 be a genus 2 surface. Its fundamental domain in the hyperbolic plane $\mathbb{H}^2$ can be chosen as an octagon P_2. The sides of P_2, denoted $e_1, e_2, \ldots, e_8$ in cyclic order, are identified in pairs by elements of the fundamental group $\Gamma_2 = \pi_1(S_2)$. A standard pairing scheme is:

- e_1 is paired with e_3^{-1} (via generator $A_1 \in \Gamma_2$).
- e_2 is paired with e_4^{-1} (via generator $B_1 \in \Gamma_2$).
- e_5 is paired with e_7^{-1} (via generator $A_2 \in \Gamma_2$).
- e_6 is paired with e_8^{-1} (via generator $B_2 \in \Gamma_2$).

These generators satisfy the relation $[A_1, B_1][A_2, B_2] = 1$ in Γ_2.

We construct a surface S_g of target genus $g \geq 2$ as an $N = (g-1)$-sheeted regular covering of S_2, with the deck transformation group being $Z_N = \{\bar{0}, \bar{1}, \ldots, \overline{N-1}\}$ (integers modulo N). This covering is defined by a surjective homomorphism $\phi : \Gamma_2 \to Z_N$. For specificity and simplicity in deriving the gluing rules, we choose the homomorphism:

$$\phi(A_1) = \bar{1}, \quad \phi(B_1) = \bar{0}, \quad \phi(A_2) = \bar{0}, \quad \phi(B_2) = \bar{0}.$$

The fundamental group of S_g is then $\Gamma_g = \ker(\phi)$.

The surface S_g can be formed by taking N copies of the octagon P_2, which we denote P_2^j for $j \in \{0, 1, \ldots, N-1\}$. Let e_i^j be the i-th edge of the j-th octagon P_2^j. The gluing rules for these $8N$ edges to form S_g are determined as follows: if an edge e_i in the original P_2 is paired with e_k^{-1} by an element $X \in \Gamma_2$, then the edge e_i^j in P_2^j is paired with the edge $(e_k^{(j+\phi(X)) \pmod N})^{-1}$ in $P_2^{(j+\phi(X)) \pmod N}$.

Applying this principle with our chosen ϕ:

1. For e_1^j (paired by A_1, with $\phi(A_1) = \bar{1}$): e_1^j is paired with $(e_3^{(j+1) \pmod N})^{-1}$.
2. For e_2^j (paired by B_1, with $\phi(B_1) = \bar{0}$): e_2^j is paired with $(e_4^j)^{-1}$.
3. For e_5^j (paired by A_2, with $\phi(A_2) = \bar{0}$): e_5^j is paired with $(e_7^j)^{-1}$.
4. For e_6^j (paired by B_2, with $\phi(B_2) = \bar{0}$): e_6^j is paired with $(e_8^j)^{-1}$.

These pairings apply for each $j \in \{0, 1, \ldots, N-1\}$.

3 Proof of Genus for the General Construction

We use the Euler characteristic formula $\chi = V - E + F$ for the cell complex formed by the N octagons and their identifications. Since all gluing rules are of the form $e_a \leftrightarrow (e_b)^{-1}$ (identifying an edge with the inverse of another), the resulting surface is orientable. For a closed orientable surface of genus g_{calc}, its Euler characteristic is $\chi = 2 - 2g_{calc}$.

1. **Number of Faces (F)**: We start with N octagons, so $F = N$. Since $N = g - 1$, we have $F = g - 1$.
2. **Number of Edges (E)**: There are $8N$ edges in total before gluing. Each of the $4N$ gluing rules (N rules for each of the four pairing types listed above) identifies two edges. Thus, the number of distinct edges after gluing is $E = \frac{8N}{2} = 4N$. So, $E = 4(g-1)$.
3. **Number of Vertices (V)**: Let v_k^j denote the k-th vertex of the j-th octagon P_2^j (e.g., v_1^j is between e_8^j and e_1^j, v_2^j is between e_1^j and e_2^j, and so on). The gluing rules of type 2, 3, and 4 (corresponding to B_1, A_2, B_2 for which $\phi(X) = \bar{0}$) occur *within* each octagon P_2^j. For a single octagon forming a genus 2 component (if its own e_1, e_3 were paired internally), all 8 vertices become equivalent. However, here e_1, e_3 are involved in inter-octagon gluing. The internal identifications within P_2^j via B_1, A_2, B_2 lead to two local vertex classes in each P_2^j:
   - $G_{jA} = \{v_1^j, v_2^j, v_5^j, v_6^j, v_7^j, v_8^j\}$ (6 vertices)
   - $G_{jB} = \{v_3^j, v_4^j\}$ (2 vertices)

The type 1 gluing rule, $e_1^j \leftrightarrow (e_3^{(j+1) \pmod N})^{-1}$, connects vertices across different octagons. Specifically, it identifies $t(e_1^j) = v_1^j$ with $h(e_3^{(j+1)}) = v_4^{(j+1)}$, and $h(e_1^j) = v_2^j$ with $t(e_3^{(j+1)}) = v_3^{(j+1)}$. Both these identifications imply that the vertex class G_{jA} from octagon j becomes equivalent to the vertex class $G_{(j+1) \pmod N,B}$ from octagon $(j+1) \pmod N$. We have N local classes of type A $(A_0, \ldots, A_{N-1})$ and N local classes of type B $(B_0, \ldots, B_{N-1})$. The N relations $A_j \sim B_{(j+1) \pmod N}$ (for $j = 0, \ldots, N-1$) link these $2N$ local classes. These relations form N disjoint sets of equivalences, each combining one A-type class with one B-type class (e.g., A_j is identified with B_{j+1}). Therefore, there are N distinct global equivalence classes of vertices. Thus, $V = N = g - 1$.

Group-Theoretic Summary of Vertex Counting. Alternatively, the number of vertices can be understood from a group-theoretic perspective. Since the map $S_g \to S_2$ is a Z_N-regular covering, defined by the surjective homomorphism $\phi : \Gamma_2 \to Z_N$ with $\Gamma_g = \ker(\phi)$, the vertices of S_g correspond to the orbits of the vertices of the N octagons (which form the fundamental domain P_g^* when appropriately arranged in $\mathbb{H}^2$) under the action of Γ_g. For a regular covering acting on the set of preimages of a point (or on the vertices of a fundamental domain for the covering group), the number of distinct orbits that form the vertices in the quotient surface S_g results from the specific identifications dictated by Γ_g. As demonstrated by the detailed pairing analysis above, this leads to exactly N distinct vertex classes in S_g.

4. **Euler Characteristic (χ)**:

   $$\chi = V - E + F = N - 4N + N = -2N$$

   Substituting $N = g - 1$:

   $$\chi = -2(g-1) = 2 - 2g$$

5. **Calculated Genus (g_{calc})**: Since the surface is orientable, $\chi = 2 - 2g_{calc}$. Equating the two expressions for χ:

   $$2 - 2g_{calc} = 2 - 2g$$

   This implies $g_{calc} = g$.

This general calculation confirms that the construction method yields an orientable closed surface of the intended genus g.

Example: Genus $g = 4$ (from $N = 3$ Octagons)

Let $g = 4$, so $N = g - 1 = 3$. We use three octagons, P_2^0 (octagon 0), P_2^1 (octagon 1), and P_2^2 (octagon 2). The gluing rules derived from the general formula are: For $j = 0$:

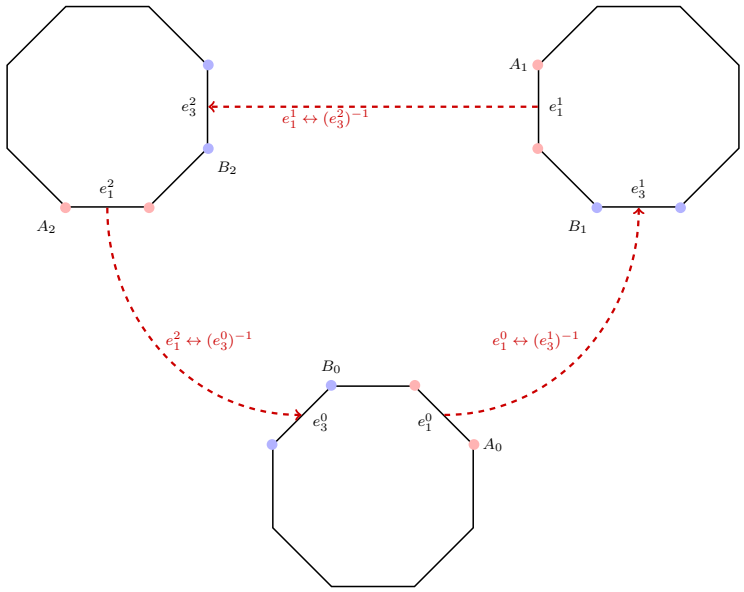


Figure 1: The octagons P_2^0, P_2^1, and P_2^2 with their edges and vertices labeled according to the gluing rules.

- $e_1^0 \leftrightarrow (e_3^{(0+1) \pmod 3})^{-1} \Rightarrow e_1^0 \leftrightarrow (e_3^1)^{-1}$
- $e_2^0 \leftrightarrow (e_4^0)^{-1}$
- $e_5^0 \leftrightarrow (e_7^0)^{-1}$
- $e_6^0 \leftrightarrow (e_8^0)^{-1}$

For $j = 1$:

- $e_1^1 \leftrightarrow (e_3^{(1+1) \pmod 3})^{-1} \Rightarrow e_1^1 \leftrightarrow (e_3^2)^{-1}$
- $e_2^1 \leftrightarrow (e_4^1)^{-1}$
- $e_5^1 \leftrightarrow (e_7^1)^{-1}$
- $e_6^1 \leftrightarrow (e_8^1)^{-1}$

For $j = 2$:

- $e_1^2 \leftrightarrow (e_3^{(2+1) \pmod 3})^{-1} \Rightarrow e_1^2 \leftrightarrow (e_3^0)^{-1}$
- $e_2^2 \leftrightarrow (e_4^2)^{-1}$
- $e_5^2 \leftrightarrow (e_7^2)^{-1}$
- $e_6^2 \leftrightarrow (e_8^2)^{-1}$

The calculation of the genus proceeds as follows:

1. **Faces (F)**: We start with $N = 3$ octagons, so $F = 3$.

2. **Edges (E)**: There are $8N = 8 \times 3 = 24$ edges in total before gluing. There are $4N = 4 \times 3 = 12$ gluing relations. Thus, the number of distinct edges after gluing is $E = \frac{24}{2} = 12$.

3. **Vertices (V)**: For each octagon P_2^j ($j = 0, 1, 2$), the internal gluings (type 2, 3, 4, involving B_1, A_2, B_2 where $\phi(X) = \bar{0}$) result in two local vertex classes:

   - $G_{jA} = \{v_1^j, v_2^j, v_5^j, v_6^j, v_7^j, v_8^j\}$
   - $G_{jB} = \{v_3^j, v_4^j\}$

   This gives $2N = 2 \times 3 = 6$ local vertex classes: $G_{0A}, G_{0B}, G_{1A}, G_{1B}, G_{2A}, G_{2B}$. The type 1 gluing rules (involving A_1 where $\phi(A_1) = \bar{1}$) connect vertices across different octagons:

   - $e_1^0 \leftrightarrow (e_3^1)^{-1}$ implies $G_{0A} \sim G_{1B}$.
   - $e_1^1 \leftrightarrow (e_3^2)^{-1}$ implies $G_{1A} \sim G_{2B}$.
   - $e_1^2 \leftrightarrow (e_3^0)^{-1}$ implies $G_{2A} \sim G_{0B}$.

   These $N = 3$ relations link the $2N = 6$ local classes into $N = 3$ distinct global vertex classes:

   - $VC_0 = G_{0A} \cup G_{1B}$
   - $VC_1 = G_{1A} \cup G_{2B}$
   - $VC_2 = G_{2A} \cup G_{0B}$

   Thus, $V = N = 3$.

4. **Euler Characteristic (χ)**:

$$\chi = V - E + F = 3 - 12 + 3 = -6$$

5. **Calculated Genus (g_{calc})**: Since the surface is orientable, $\chi = 2 - 2g_{calc}$.

$$-6 = 2 - 2g_{calc} \implies 2g_{calc} = 8 \implies g_{calc} = 4$$

This matches the intended genus $g = 4$.

4 Geometric Realization, Covering Structure, and Fundamental Group Hierarchy

This section aims to provide a deeper understanding of the key features of the proposed rotational construction for genus g surfaces. We will elaborate on how this construction naturally gives rise to a covering map, its geometric realizability in hyperbolic space, and the profound algebraic relationships it reveals among fundamental groups. The method detailed offers an alternative to the standard procedure of building a genus g surface from a single $4g$-sided polygon, particularly emphasizing a hierarchical relationship and inherent symmetries.

4.1 The Covering Map $p : S_g \to S_2$ and its Properties

The core of our construction lies in viewing the genus g surface S_g as an $N = (g-1)$-sheeted regular covering of a genus 2 surface S_2, with the cyclic group Z_N acting as the group of deck transformations.

Definition and Deck Transformations. The construction inherently defines an $N = (g-1)$-sheeted regular covering map $p : S_g \to S_2$. The deck transformation group is isomorphic to $Z_N = \{\bar{0}, \bar{1}, \ldots, \overline{N-1}\}$. An element $\bar{k} \in Z_N$ acts by mapping the j-th octagonal sheet P_2^j to $P_2^{(j+k) \pmod N}$, while preserving the gluing rules. For any point $x \in S_2$, its preimage $p^{-1}(x)$ in S_g consists of N distinct points, one in each octagonal sheet P_2^j. These N octagons, when assembled according to the gluing rules, form a fundamental domain for S_g, and each sheet P_2^j projects down to S_2 via p, effectively "covering" the base surface S_2.

Monodromy. The monodromy action of $\Gamma_2 = \pi_1(S_2)$ on a fiber $p^{-1}(x)$ is determined by the homomorphism $\phi : \Gamma_2 \to Z_N$. For a loop $\gamma \in \Gamma_2$ based at $x_0 \in S_2$, lifting γ to a path in S_g starting at a point $y_j \in p^{-1}(x_0)$ (where y_j is in the j-th sheet) leads to an endpoint $y_{(j+\phi(\gamma)) \pmod N}$. Thus, a loop γ with $\phi(\gamma) = \bar{k}$ induces a cyclic permutation of the sheets by k.

4.2 Fundamental Group Hierarchy and Algebraic Structure

The covering $p : S_g \to S_2$ establishes a clear hierarchical relationship between the fundamental groups of S_g and S_2.

The Exact Sequence. The fundamental group of the constructed surface S_g, denoted $\Gamma_g = \pi_1(S_g)$, is a normal subgroup of $\Gamma_2 = \pi_1(S_2)$. The quotient group Γ_2/Γ_g is isomorphic to the deck transformation group Z_N. This relationship is captured by the short exact sequence:

$$1 \to \Gamma_g \to \Gamma_2 \xrightarrow{\phi} Z_N \to 1.$$

Here, $\Gamma_g = \ker(\phi)$, and its index in Γ_2 is $N = g - 1$.

Generators of Γ_g. The group Γ_g is generated by lifts of those loops in S_2 whose image under ϕ is the identity element $\bar{0} \in Z_N$.

- Loops in S_2 corresponding to the generators B_1, A_2, B_2 (for which $\phi(B_1) = \phi(A_2) = \phi(B_2) = \bar{0}$) lift to closed loops within each sheet of S_g. These lifts are elements of Γ_g.
- A loop in S_2 corresponding to the generator A_1 (for which $\phi(A_1) = \bar{1}$) lifts to a path in S_g connecting a point in sheet P_2^j to a corresponding point in sheet $P_2^{(j+1) \pmod N}$. Iterating such a loop N times (i.e., A_1^N) results in a loop in S_2 whose image $\phi(A_1^N) = N \cdot \bar{1} = \bar{0}$ in Z_N. Thus, A_1^N lifts to a closed loop in S_g and is an element of Γ_g. Similarly, elements like $A_1 B_1 A_1^{-1}$ (if $\phi(B_1) \neq \bar{0}$ in a different homomorphism choice) would need to be considered based on their image under ϕ.

The Special Role of Γ_2. In the context of this construction, where S_g is built as a specific Z_{g-1}-symmetric cover of a fixed S_2, the group Γ_2 acts as a parent group from which a family of groups $\{\Gamma_g\}_{g \geq 2}$ is derived. More broadly, in the lattice of fundamental groups of closed orientable hyperbolic surfaces ordered by subgroup inclusion, Γ_k (for genus $k \geq 2$), groups Γ_2 are maximal elements, as they cannot be proper finite-index subgroups of another Γ_h (where $h \geq 2$).

4.3 The Fundamental Domain P_g^* and Hyperbolic Geometric Realization

The union of the $N = (g-1)$ octagons in the hyperbolic plane, $P_g^* = \bigcup_{j=0}^{N-1} \alpha_j(P_2)$ (where P_2 is a fundamental octagon for $\Gamma_2 = \pi_1(S_2)$, and α_j are appropriately chosen isometries corresponding to the sheets), forms a legitimate fundamental domain for the resulting genus g surface $S_g = \mathbb{H}^2/\Gamma_g$. The gluing rules defined are precisely the side-pairings of this larger domain P_g^* by elements of $\Gamma_g = \ker(\phi)$.

Hyperbolic Realizability. The geometric realization of this construction can be grounded in hyperbolic geometry. We can choose P_2 to be a regular octagon in the hyperbolic plane $\mathbb{H}^2$ with each internal angle being $\pi/4$. Such an octagon exists (as $8 \times \pi/4 = 2\pi < (8-2)\pi = 6\pi$). When its 8 sides are paired according to the standard gluing for a genus 2 surface (e.g., $e_1 \leftrightarrow e_3^{-1}, e_2 \leftrightarrow e_4^{-1}, e_5 \leftrightarrow e_7^{-1}, e_6 \leftrightarrow e_8^{-1}$), all 8 vertices of P_2 are identified to a single point in S_2. The sum of the angles around this point is $8 \times (\pi/4) = 2\pi$, satisfying the angle condition of Poincaré's polygon theorem for P_2 to be a fundamental domain for S_2. The Bolza surface is a prime example of a genus 2 surface admitting such a regular octagonal fundamental domain.

Now, consider the fundamental domain $P_g^* = \bigcup_{j=0}^{N-1} P_2^j$ formed by N such identical regular octagons P_2^j, arranged and glued according to our Z_N-symmetric

rules. As established in our vertex analysis (Section 3, item 3), the $8N$ initial vertices are grouped into N distinct equivalence classes in S_g. Each such vertex class in S_g is formed by exactly 8 original "corners" from the constituent octagons. If each of these 8 corners contributes an angle of $\pi/4$, the total angle sum around each of the N distinct vertices in S_g is $8 \times (\pi/4) = 2\pi$. This satisfies the angle condition of Poincaré's polygon theorem for P_g^* to be a fundamental domain for $S_g = \mathbb{H}^2/\Gamma_g$. Thus, the constructed surface S_g can be realized as a hyperbolic surface with a metric derived from this tessellation.

Relation to the $4g$-gon. While the canonical fundamental domain is often presented as a $4g$-gon, P_g^* is a polygon (formed by N adjoined octagons) whose boundary edges are identified by Γ_g to yield S_g. P_g^* can be transformed into a standard $4g$-gon via topological cutting and pasting operations.

Holonomy Representation. When P_2 is chosen as a regular hyperbolic octagon (like that for the Bolza surface), the monodromy action described earlier directly corresponds to the holonomy representation of the resulting hyperbolic structure on S_g. The deck transformations Z_N manifest as isometries of S_g.

4.4 Flexibility and Generalizations of the Construction

The specific choice of homomorphism ϕ and the nature of the quotient group Z_N offer avenues for variations and generalizations.

Dependence on the Homomorphism ϕ. Our chosen homomorphism $\phi(A_1) = \bar{1}, \phi(B_1) = \phi(A_2) = \phi(B_2) = \bar{0}$ is not unique. Different surjective homomorphisms $\phi' : \Gamma_2 \to Z_N$ will generally lead to different covering spaces S'_g.

- For instance, if one chose $\phi'(B_1) = \bar{1}$ and $\phi'(A_1) = \phi'(A_2) = \phi'(B_2) = \bar{0}$, the gluing rule for edge e_2^j would become $e_2^j \leftrightarrow (e_4^{(j+1) \pmod N})^{-1}$. This would alter the vertex identification cycles and potentially the geometric properties of the resulting surface S'_g (e.g., its full isometry group), even though its genus, as determined by the Euler characteristic calculation based on N octagons, would remain $g = N + 1$.
- The genus of the resulting surface is invariant under such choices (as long as ϕ is surjective and $N = g - 1$), but geometric features, such as the lengths of shortest geodesics or the full automorphism group of S_g, may differ.

Generalizations to Non-Cyclic Covers. While this note has focused on cyclic covers (Z_N), the general principle of constructing S_g from N copies of P_2 can be extended to situations where Γ_g is a normal subgroup of Γ_2 such that $\Gamma_2/\Gamma_g \cong G$, where G is any finite group of order N.

- If G is a non-abelian group, the covering $S_g \to S_2$ would be regular but not necessarily cyclic. The N sheets would correspond to the elements of G, and the gluing rule for an edge e_i^x (in sheet $x \in G$) paired by $X \in \Gamma_2$ would be $e_i^x \leftrightarrow (e_k^{x \cdot \phi(X)})^{-1}$ (or $e_k^{\phi(X) \cdot x}$ depending on convention).
- The genus calculation via Euler characteristic remains valid ($F = N$, $E = 4N$, leading to $V = N$ for $\chi = 2 - 2N = 2 - 2g$), but the specific vertex and edge identification patterns would be more complex, reflecting the structure of G.

This highlights that S_2 can serve as a base for constructing a rich variety of higher genus surfaces with diverse symmetry groups.

5 Conclusion

This note presents a novel rotational construction of genus g surfaces from $(g-1)$ octagons, leveraging the hierarchical relationship between surfaces of different genera. By interpreting S_g as a Z_N-symmetric covering of a genus 2 surface S_2, we derive explicit gluing rules for assembling S_g from N octagonal fundamental domains. Key contributions include:

1. **Constructive Framework**: A systematic method to build S_g via cyclic covers, validated by Euler characteristic calculations.
2. **Geometric Realization**: Demonstration that S_g admits a hyperbolic metric when the base octagon P_2 is regular (e.g., Bolza surface), satisfying Poincaré's polygon theorem.
3. **Fundamental Group Insights**: Clarification of the subgroup hierarchy $\Gamma_g \triangleleft \Gamma_2$ and the role of homomorphisms $\phi : \Gamma_2 \to Z_N$ in defining covers.

This approach offers pedagogical and practical advantages:

- **Visualization**: The cyclic gluing of octagons provides an intuitive model for higher-genus surfaces.
- **Hierarchy Emphasis**: S_2 serves as a "building block" for symmetric S_g, contrasting with the standard $4g$-gon construction.
- **Computational Utility**: The method simplifies genus calculations and generalizes to irregular covers.

Future work could explore:

- **Alternative Homomorphisms**: Classifying covers arising from different ϕ and their geometric properties.
- **Higher Symmetries**: Extending the construction to surfaces with larger automorphism groups (e.g., Hurwitz surfaces).

- **Applications**: Using this framework to study spectral theory or heat kernels on symmetric hyperbolic surfaces.

In summary, this rotational construction enriches the tapestry of surface topology, highlighting the genus 2 surface as a foundational element in the architecture of higher-genus spaces.